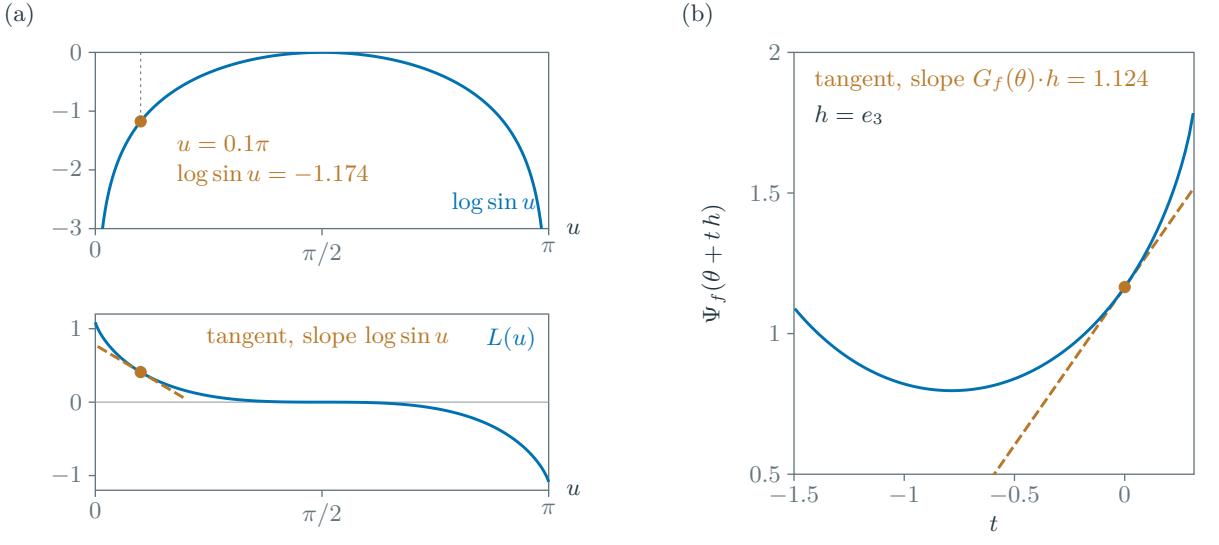

(a)
0
−1
−2
−3
$u = 0.1\pi$
$\log\sin u = -1.174$
$\log\sin u$
0
$\pi/2$
π
u
1
0
−1
tangent, slope $\log\sin u$
$L(u)$
0
$\pi/2$
π
u
(b)
2
1.5
1
0.5
tangent, slope $G_f(\theta)\cdot h = 1.124$
$h = e_3$
$\Psi_f(\theta + t\,h)$
−1.5
−1
−0.5
0
t